Design and Verification of Flip-flop

Objectives:

1. To design and implement basic flip flops circuits.
2. To verify the functional behaviour and truth table of the implemented flip-flop through experimental observation.

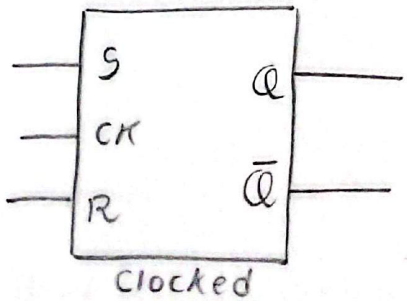
Basic Theory:

A flip-flop is a basic digital memory element that can store one bit of data. It is a type of bistable circuit, meaning it has two stable states: 0 (reset) and 1 (set). Unlike combinational circuits, flip-flops have memory - the output depends not just on current inputs but also on the previous state.

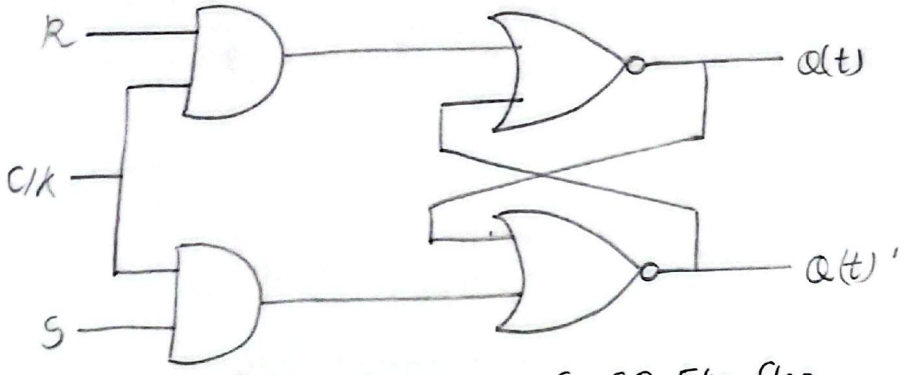
There are mainly four types of flip-flops:

1. SR Flip-Flop (Set-Reset):

It has two inputs: S (Set) and R (Reset). When S=0, R=0, Output stays the same. When S=0, R=1, output Q goes to 0. The condition S=R=1 is not allowed (invalid State).



Block diagram of SR Flip flop.

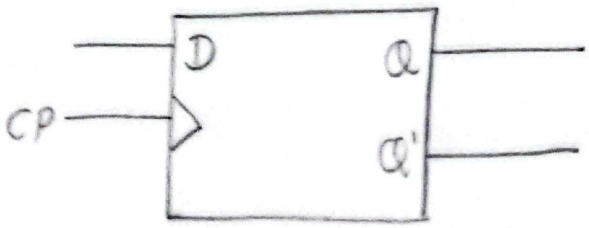


Logic diagram of SR Flip flop

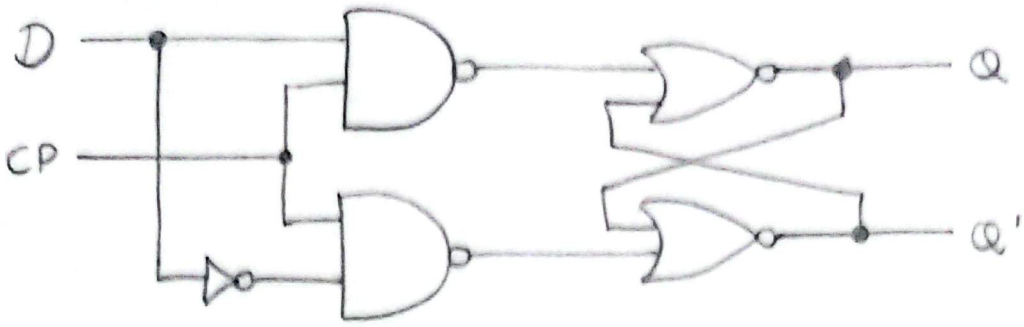
Q(t)	S	R	Q(t+1)
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	In determinate
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	Indeterminate

2. D Flip-flop:

It has one data input (D). Whatever value is present at D gets stored in Q at the clock edge. It removes the invalid state problem of SR flip-flop by connecting S = D and R = D'.



Block diagram of D flip-flop



Q(t)	D	Q(t+1)
0	0	0
0	1	1
1	0	0
1	1	1

fig. Logic Diagram and Truth Table of D flip flop

3. JK Flip-flop:

It is an improved version of SR flip-flop. J works like S(set) and K works like R(reset). When J=K=1, the output toggles (Flips). This removes the invalid state of SR flip-flop.

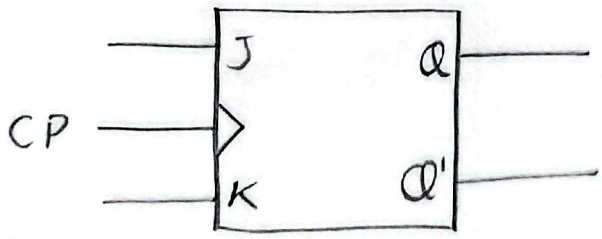


fig. Block Diagram of JK-flip flop

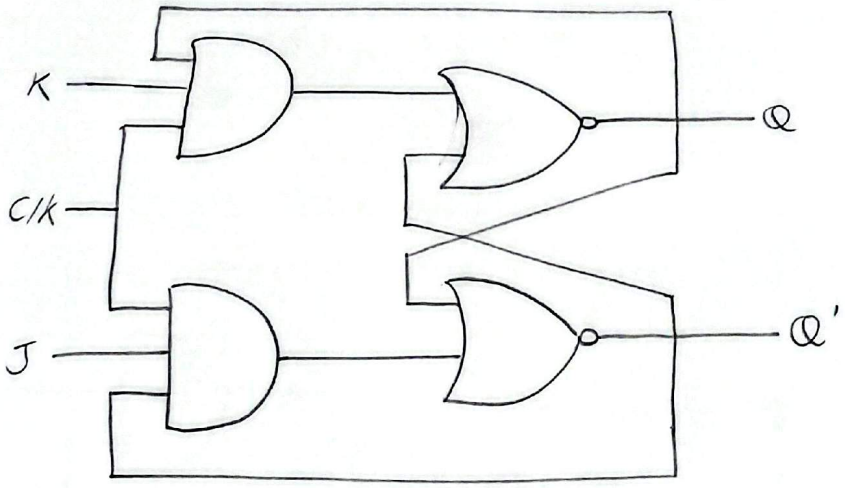


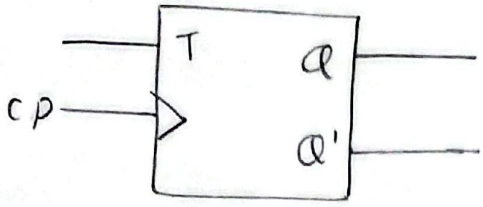
fig: Logic diagram of JK flip flop

Q(t)	J	K	Q (t+1)
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

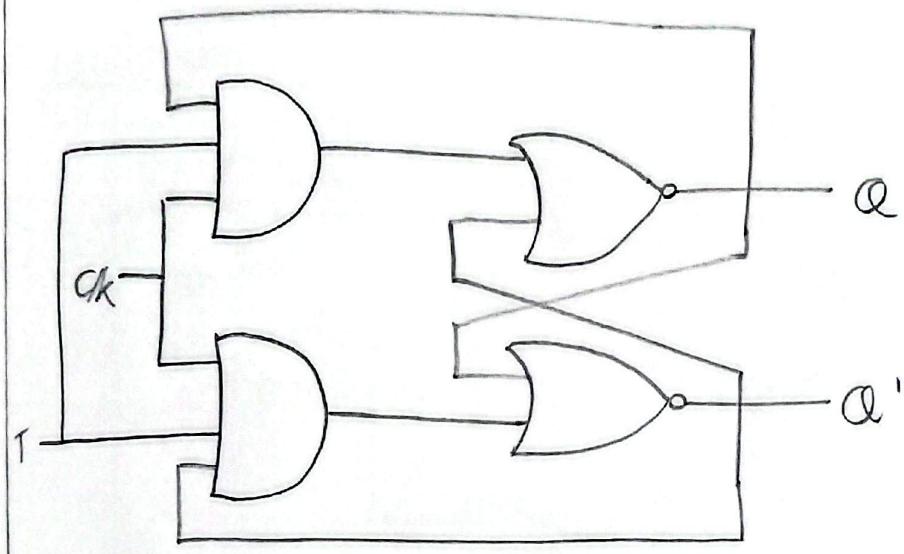
Truth Table of JK flip flop

4. T- flip flop:

It has one input T. When T=0, output stays the same. When T=1, output toggles. It is made from JK flip-flop by connecting J=K=T.



Block diagram



Logic diagram

Truth Table

Q(t)	T	Q(t+1)
0	0	0
0	1	1
1	0	1
1	1	0

fig. T flip-flop.

Diagrams and Observation:

Objective 1: Building a SR flip flop using AND and NOR gate

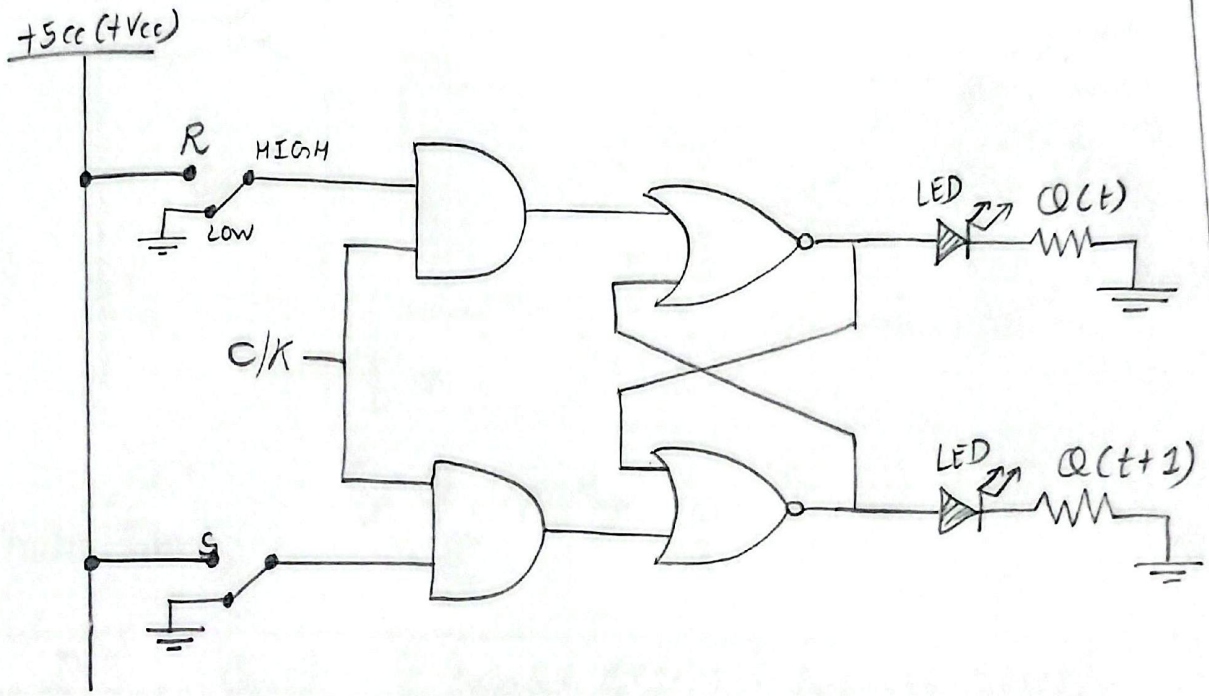


fig. SR flip flop

Q(t)	S	R	Expected Q(t+1)	Observed Q(t+1)
0	0	0	0	0
0	0	1	0	0
0	1	0	1	1
0	1	1	X	X
1	0	0	1	1
1	0	1	0	0
1	1	0	1	1
1	1	1	X	X

Hence, The observed results and expected results are the same.

Observation 2: Building a D Flip-flop using NAND, NOT, NOR gate

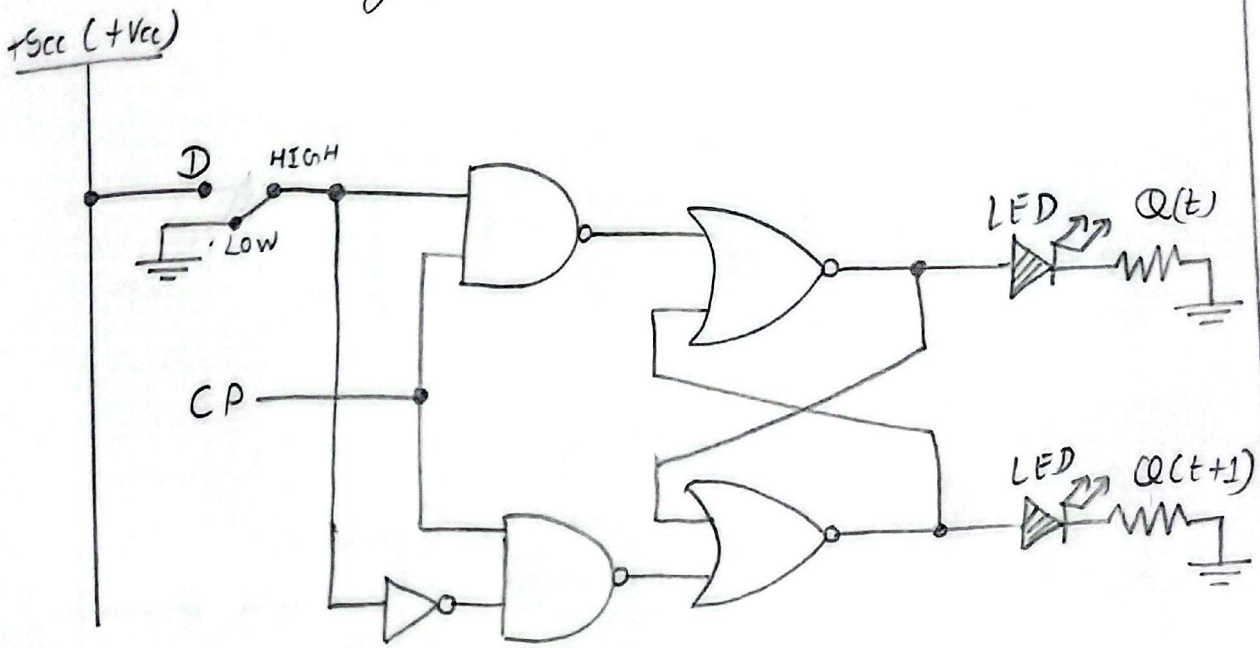


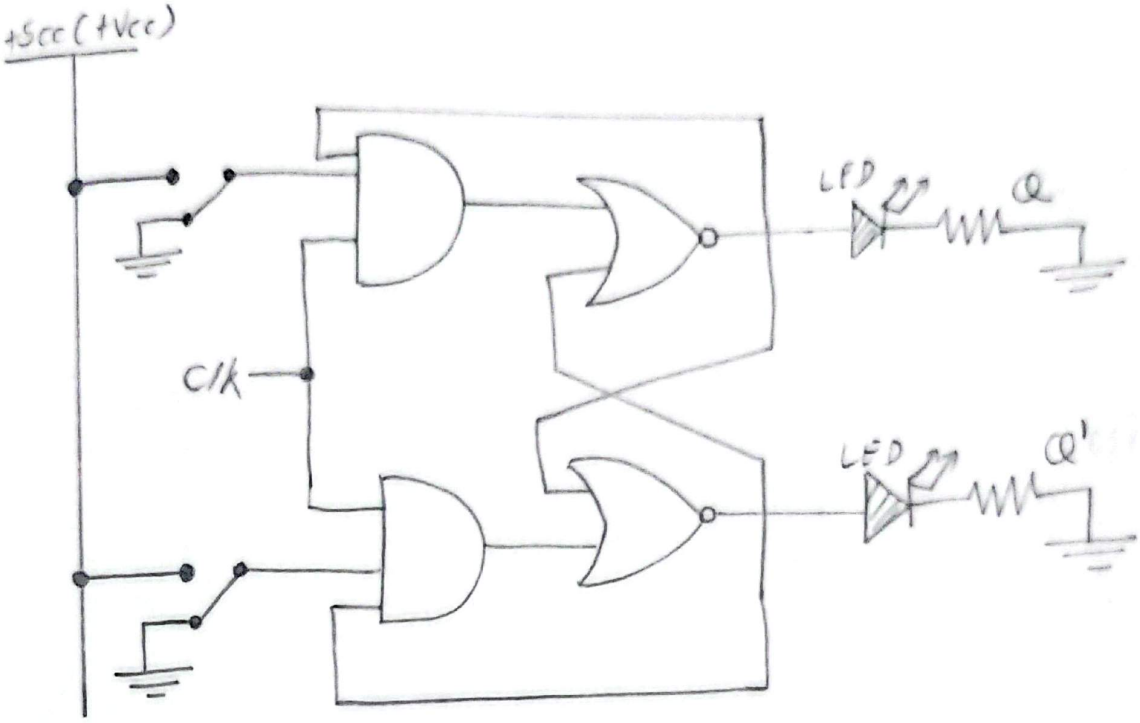
fig. D flip-flop

Truth Table

D	Q(t)	Expected Q(t+1)	Observed Q(t+1)
0	0	0	0
0	1	0	0
1	0	1	1
1	1	1	1

Hence, The Expected results and observed result are the same.

Objective 3: Building a JK flip flop using AND and NOR Gate

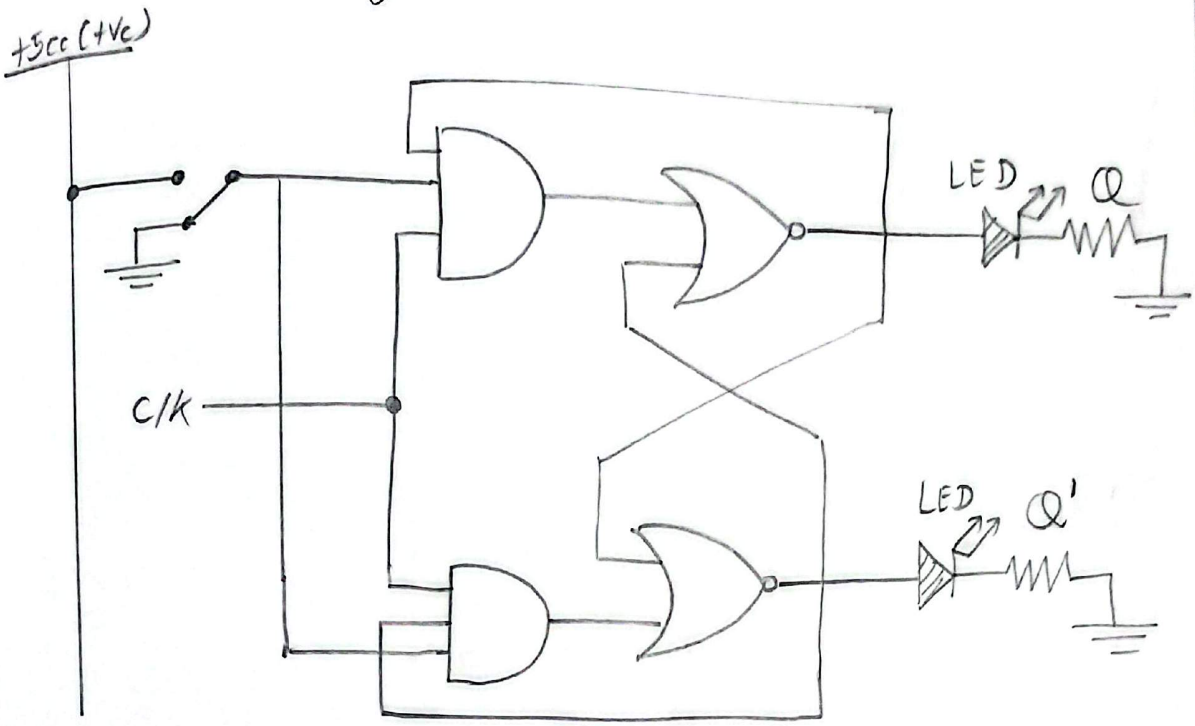


Truth Table

Q(t)	J	K	Expected Q(t+1)	Observed Q(t+1)
0	0	0	0	0
0	0	1	0	0
0	1	0	1	1
0	1	1	1	1
1	0	0	1	1
1	0	1	0	0
1	1	0	1	1
1	1	1	0	0

Hence, The Expected result and Observed result are the same.

Objective 4 : Building a T flip-flop using AND and NOR gate



Truth Table

Q(t)	T	Expected Q(t+1)	Observed Q(t+1)
0	0	0	0
0	1	1	1
1	0	1	1
1	1	0	0

Hence, The Expected result and Observed result are the same.

Discussion and Conclusion :

In this lab, we designed and implemented four basic flip-flop circuits. We verified their truth tables by applying different input combinations and observing the output.